

Data Card: Zimbabwe Agricultural Extension RAG Dataset

Input vs. Output • What are users inputting (e.g., audio, text, images)? • What are the outputs or labels your model will predict or generate?	Users input natural language text questions from smallholder farmers in Zimbabwe, covering crop diseases, pests, nutrient deficiencies, soil management, climate adaptation, and fertiliser application. Inputs are primarily in English, but may also include expressions from local languages such as Shona or Ndebele. The output is a ranked list of documents (top 5) from the agricultural extension knowledge base, containing the most relevant agronomic advice, disease control measures, and fertiliser recommendations. The model performs a retrieval task — it does not generate new text, but returns existing documents ranked by relevance to the query.
Classification Problem • What is the classification task, if any? • What categories or classes are you working with, and why?	This is an information retrieval / ranking task, not a traditional classification task. The model computes relevance scores between each document in the knowledge base and the user's question, then ranks the documents accordingly. Documents are grouped into the following agricultural themes: • Crop diseases • Pests • Nutrient deficiencies • Soil management • Climate adaptation • Fertiliser advice These categories reflect the most common agricultural challenges faced by smallholder farmers in Zimbabwe.
Labeling • Where do your labels come from — domain experts, community members, crowd workers, AI-generated sources? • How will you check for consistency and accuracy?	Relevance labels come from human expert judgments provided by agricultural domain specialists — such as extension officers from Zimbabwe's AGRITEX (Department of Agricultural, Technical and Extension Services), agronomists, and crop scientists. Each (question, document) pair is independently rated for relevance by multiple experts to compute the nDCG@5 metric. • Multi-expert independent labeling with inter-rater agreement (IRA) measured • Disagreements resolved through expert adjudication and consensus discussion • A golden query set is maintained for ongoing validation of retrieval quality
Data Creation • How will you collect your data? • Will it come from direct sources (like interviews or recordings), or will you synthesize data in low-resource settings?	Data will be sourced from: • Existing agricultural extension materials published by Zimbabwe's Ministry of Lands, Agriculture, Fisheries, Water and Rural Development, AGRITEX, and CGIAR • Localised knowledge resources tailored to Zimbabwe's Natural Regions (farming zones), including crop guides, livestock disease bulletins, and soil fertility manuals • Publicly available agricultural corpora, such as the CGIAR research repository (45,232 publications) We will primarily use existing documented resources. In low-resource settings, we may:

	• Partner with AGRITEX to transcribe and digitise printed extension materials • Localise and adapt international agricultural knowledge to the Zimbabwean context • Avoid synthetic data to ensure factual accuracy and practical relevance
Web Collection Ethics • Do you plan to source data from the web? If yes, which sites or platforms? • What tools will you use — AI tools, human workers, or both? • Have you addressed issues of consent, copyright, and privacy?	Yes. We plan to collect publicly available agricultural documents from: • CGIAR open-access publications • CABI Plant Health Knowledge Bank • Kaggle existing agricultural datasets • Zimbabwe government agricultural websites Tools: • Automated web scraping for publicly available documents • Human screening to assess document quality and Zimbabwe-specific relevance • Domain experts to validate content accuracy and local applicability Consent, copyright, and privacy: • Only open-access / publicly licensed documents are used • All sources are attributed with licence information (e.g., CC-BY) • No personally identifiable information (PII) is collected • Every document is tagged with its source and version for traceability
Bias and Representation • Who might be left out of your dataset? • How will you ensure that the dataset reflects the diversity of the community and includes underrepresented voices?	• Non-English speakers: Most smallholder farmers in Zimbabwe speak Shona or Ndebele; an English-only dataset may exclude them • Farmers in remote areas: Their unique practices and challenges may not be captured in formal written literature • Women farmers: Agricultural extension literature may underrepresent their perspectives and priorities • Collaborate with AGRITEX extension officers to capture local knowledge that reflects real on-the-ground realities • Include local-language question–document pairs where feasible • Consult community representatives (including women's farming groups) to validate content relevance • Document the dataset's geographic coverage and known limitations transparently
Values and Community Alignment • How does your dataset design reflect the values you identified in 01 Build Your Own Small Language Model? • How will your community benefit from this dataset, and how will you ensure accountability to them?	Our dataset design is grounded in the following values: • Equitable access: Providing free, offline-capable agricultural knowledge to resource-constrained farmers • Knowledge sovereignty: Prioritising Zimbabwean local agricultural knowledge over externally imposed standards • Sustainability: Supporting climate-resilient farming and food security • Inclusivity: Striving to cover local languages and marginalised groups Direct benefits:

	• Farmers receive timely, accurate advice — reducing dependence on scarce extension workers Indirect benefits: • Improved crop yields, better food security, and enhanced climate resilience Accountability mechanisms: • Establish feedback channels with AGRITEX and farmer cooperatives • Regularly evaluate retrieval outputs for practical usefulness and accuracy • Publish this Data Card publicly to document limitations, assumptions, and known biases • Allow community members to challenge and supplement knowledge base content
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Dataset Summary

The Zimbabwe Agricultural Extension RAG Dataset is a retrieval knowledge base designed for a Retrieval-Augmented Generation (RAG) system serving smallholder farmers in Zimbabwe. It contains curated agricultural documents from AGRITEX, CGIAR, CABI, and other sources, covering crop diseases, pests, nutrient deficiencies, soil management, climate adaptation, and fertiliser advice. All documents are human-screened and localised to the Zimbabwean agricultural context. The dataset is used to train and evaluate document retrieval models, with top-5 ranking performance measured by nDCG@5. The dataset is built transparently and responsibly, with full documentation of sources, labeling methods, and limitations.

This Data Card follows the Data Cards Playbook framework to ensure transparent and accountable dataset documentation.